

From Set-Level Inferability to Field-Level Risk: Budgeted Marginal Inference and an Amortized Attacker for Grading Power Market Data

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Abstract

Electricity-market data are graded and released field by field, yet the inference risk of a field depends on what else an adversary knows: fields that reveal nothing alone can, together with other released data, expose unit outputs, private offers, line flows or load before release. We define field-level risk from set-level inferability, the out-of-sample accuracy that attackers retrained on a set of fields reach for a sensitive target. The contextual gain of a field over a background has a whole distribution; attribution methods average it, whereas a security audit needs its worst case. The budgeted marginal risk $M^{(K)}$ is the largest gain over any background of at most K fields, relative to data already public, and its excess over single-field risk is the background amplification $\Gamma^{(K)}$. $M^{(K)}$ is a budget-constrained marginal contribution index, the minimal threshold-free safety margin, and flags exactly the members of minimal unsafe combinations of at most $K + 1$ fields; its top backgrounds tell the data owner what not to co-release. To evaluate all small sets repeatedly, an amortized attacker—one network pre-trained under random visibility masks, queried through a closed-form readout per field set—scans the sets, and retraining a few backgrounds per field certifies the result. In a controlled market case where one unit’s private offer markup is the target, every price field has zero risk alone and all of its risk comes from the unit’s output as background, ranked and regime-dependent as the dispatch physics implies; correlation, single-field and attribution scores miss it. On four power data sets with rule-based field roles and 1.2×10^5 exhaustively retrained field sets as ground truth, amplification is common—a typical field’s average contextual gain is a third of its worst case—and single-field grading recovers only 13–17% of the critical fields, whereas scan–certify

recovers up to 88% without false positives. We also find that masked pre-training on the targets alone collapses the representation and, on the small sets of an audit, adds little over random features; masked reconstruction of all columns with a safeguard against exploding predictions improves every accuracy metric.

Keywords: data classification and grading, inference attack, electricity market disclosure, feature importance, marginal contribution, amortized inference

1. Introduction

Electricity markets publish a large and growing amount of operational data. Disclosure rules list, field by field, what must be released and when: forecasts and requirements before clearing, prices and their components right after clearing, and actual operating quantities on the following day [1]. Other quantities—the output or the offers of an individual generating unit, the hourly flow on a specific line, or the actual load before it is disclosed—are protected, either permanently or until a later release date. Data classification standards ask operators to grade each field and, explicitly, to consider how aggregation and processing change a field’s level [2]. How to do the latter is left open.

The difficulty is that the inference risk of a field is not a property of the field alone. Power data come from one physical and economic system, and the information a field adds depends on what else is known. A unit’s hourly output says little about its private offer, and neither does the price at its bus; together, whenever the unit is marginal, they reveal the offer (Section 7). The same price adds nothing over one background and a great deal over another. This is the inference and aggregation problem of database security [3, 4], and it is not hypothetical for power data: public prices alone can expose grid topology [5]. Recent work on data sensitivity assessment has recognized that a field’s risk depends on its inference relations to sensitive data and propagates risk along an inference graph [6]. What is still missing is a field-level risk that is *defined* by what an adversary can infer from a set of fields—so that it can be checked by an attack experiment rather than calibrated to expert scores—and that captures inference requiring several fields jointly.

We start from the gain $\Delta_i(T)$ that releasing field i brings over a background T of fields the adversary already holds, measured as the out-of-sample accuracy of retrained attackers. A field then has a whole distribution of contextual gains, and grading it means summarizing that distribution. Attribution methods such as Shapley values and SAGE average the gains over backgrounds; this is the right answer to “how much does the field contribute typically”, but a security audit asks whether *some* admissible background makes the field dangerous. We therefore take the maximum—the Marginal Contribution Index [7], in line with worst-case measures of leakage in quantitative information flow [8, 9]—and add the two conditions a disclosure audit needs: the data already public and a bound on the adversary’s background. The resulting budgeted marginal risk $M_{i \rightarrow y}^{(K)}$, together with its single-field part $M^{(0)}$, the *background amplification* $\Gamma^{(K)} = M_{i \rightarrow y}^{(K)} - M^{(0)}$ and the top backgrounds that attain it, is the field-level output of the audit.

Computing it requires the inferability of every small field set, and it has to be recomputed whenever targets or public data change. We amortize this with one network pre-trained under random visibility masks—an adversary that does not know in advance which fields it will obtain—and a closed-form readout solved per field set; the estimate is only used to scan, and a handful of backgrounds per field is retrained to certify the result. We found that pre-training on the sensitive targets alone contributes almost nothing beyond random features, because the representation collapses onto a few target directions, and that adding masked reconstruction of all columns, together with a safeguard against exploding predictions, improves every accuracy metric.

Contributions.

- *Background-aware field risk.* From set-level inferability we define the budgeted marginal risk $M_{i \rightarrow y}^{(K)}$ and the background amplification $\Gamma^{(K)}$, relate them to the MCI, give a closed form of the amplification at $K = 1$, and show that $M_{i \rightarrow y}^{(K)}$ is the minimal threshold-free safety margin and that its critical fields are exactly the members of small minimal unsafe sets (Section 4).
- *An amortized attacker with scan-certify.* Masked pre-training plus a per-set closed-form readout estimates V_y for every small set in milliseconds; an error analysis explains why field risk is estimated more accurately than the set values, certification by retraining gives lower

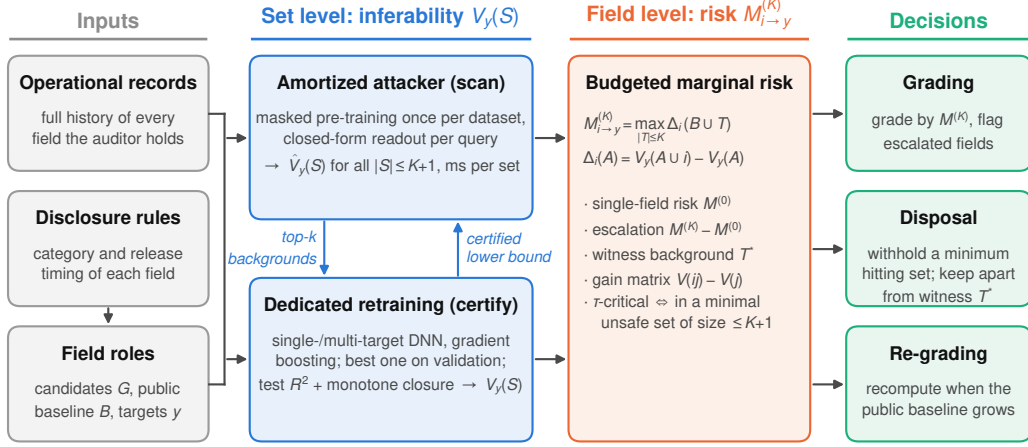


Figure 1: Audit framework. Disclosure rules fix field roles. Set-level inferability $V_y(S)$ is defined by dedicated retraining and scanned by an amortized attacker; the scan proposes, for every field, backgrounds that are certified by retraining. The field-level profile—single-field risk, background amplification and top backgrounds—supports grading and downstream disposal.

bounds that do not depend on the estimate, and a diagnosis of the pre-training shows why a target-only objective collapses and how reconstruction repairs it (Sections 5 and 10).

- *Mechanism and real-data evidence.* A controlled power-system case with a known inference mechanism, and four data sets with rule-based field roles (RTS-GMLC, the Australian NEM with actual unit outputs, PJM, CAISO) with 1.2×10^5 exhaustively retrained field sets as ground truth. We show that background amplification is common and physically interpretable, compare the definition with correlation-, attribution- and inference-graph-based scores, ablate the estimator, and test robustness and the choice of K (Sections 7–11); disposal is discussed as an application (Section 12).

2. Related Work

Inference and aggregation in data release. That individually innocuous data items can jointly reveal protected information is an old observation in database security, where it is known as the inference and aggregation problem [3, 4];

the privacy literature supplies the best-known illustration, three demographic attributes that jointly single out most individuals [10]. In power systems, public market data have been shown to expose non-public physical information, for example grid topology from locational marginal prices [5]. Privacy mechanisms for metering data [11, 12] and formal models such as differential privacy [13] or k -anonymity [14] bound what can be learned about an individual record after a release mechanism is applied. They answer a different question from the one faced by a market operator, who must decide which aggregate operational *fields* may be published at all.

Data sensitivity assessment under inference leakage. Data classification and grading is performed field by field in practice, and the Chinese national standard explicitly asks that aggregation and processing effects be considered [2], without prescribing how. The closest work to ours is IGNN [6], which introduced data inference leakage into sensitivity assessment for cyber-physical power grids: fields are nodes of an inference graph whose edges encode physical or statistical inference relations, and sensitivity is propagated along these edges by a graph attention network from an intrinsic (expert-assigned) score. This established that the risk of a field depends on its inference relations to sensitive data. Two questions remain open in this line of work. First, the risk score is calibrated against expert labels, while its ultimate validation is an attack experiment in which a model is trained on a group of fields; the attack outcome itself is not the definition. Second, an edge connects two fields, whereas inference frequently requires several fields jointly, with none of them informative alone. We take the attack outcome on a field *set* as the definition of risk and derive the field-level quantity from it.

Feature importance. Turning set-level values into per-feature scores is the subject of feature attribution. Shapley-based measures [15, 16, 17, 18] average a feature’s marginal contribution over all contexts, which is the right answer for fair allocation but dilutes the credit of redundant features [19, 7]. Leave-one-covariate-out (LOCO) [20] and permutation importance [21, 22, 23] evaluate a single context, the full feature set, so a feature with a near-substitute receives almost no importance. The Marginal Contribution Index (MCI) [7] takes the *maximum* marginal contribution over contexts, is characterized axiomatically, and is immune to dilution. Security reasoning favours the worst case for the same reason quantitative information flow replaced Shannon entropy by min-entropy and gain functions [8, 9]: an average over

contexts is not a guarantee. Our field-level risk is a budget-constrained MCI evaluated relative to already-public data.

Fast estimation of many refitted models. Computing the performance of a model refitted on many feature subsets is expensive, which has motivated lazy [24] and warm-started [25] estimators of variable importance, amortized explanation models [26, 27], and conditional generative models trained under random masks [28, 29, 30, 31]. Closed-form ridge heads on frozen features are the standard fast-adaptation device in meta-learning [32, 33], and the benefit of sharing a representation across tasks is well understood [34, 35, 36]. We combine these ingredients into an estimator of set-level inferability and, more importantly, use it inside a scan–certify loop so that decisions rest on retrained attackers rather than on the estimate.

3. Problem Formulation

3.1. Parties, field roles and threat model

The **auditor** is the data owner. It holds the complete history of all operational fields, including those it regards as sensitive, and grades fields offline before release. The **adversary** obtains published fields and is assumed to hold historical auxiliary labels of the sensitive target, from which it trains a supervised inference model; risk is therefore measured as out-of-sample *supervised* inferability. The adversary may combine the field under review with other fields it already holds.

Field roles are fixed by disclosure rules, not by data analysis. Let G be the candidate fields that the rules allow to be published, $B \subseteq G$ a *public baseline* that is already released ($B = \emptyset$ unless stated otherwise), $H = G \setminus B$ the fields under review, and Y_y a sensitive target. Section 6.1 instantiates these roles for four data sets from the categories and release timings of the Chinese electricity-market disclosure rules [1]. An ex-ante estimate of the *same physical quantity* as a target (e.g. a load forecast when the target is metered load) is removed from G , because it is a copy of the target rather than an inference channel.

3.2. Set-level inferability

Definition 1 (Population inferability).

$$V_y^*(S) = 1 - \inf_f \mathbb{E}[(Y_y - f(X_S))^2] / \text{Var}(Y_y),$$

the infimum taken over measurable f . It satisfies $V_y^*(\emptyset) = 0$, $0 \leq V_y^* \leq 1$ and is monotone in S .

Definition 2 (Empirical inferability). *The data are split into training and test rows, with a validation part inside training. For an attacker family $\mathcal{H}$, every $h \in \mathcal{H}$ is trained on X_S only; the attacker and its training epoch are selected on validation, and the test R^2 is reported: $V_y^{\text{ret}}(S) = R_{\text{test}}^2(\hat{h}_S)$ with $\hat{h}_S = \arg \max_{h \in \mathcal{H}, \text{epoch}} R_{\text{val}}^2(h; S)$.*

Finite samples, optimization and selection noise make V_y^{ret} slightly non-monotone. An adversary holding S can train on any subset of S and keep the best, so the operational object is the *monotone closure*

$$V_y(S) = V_y^{\text{ret}}(\hat{T}_S), \quad \hat{T}_S = \arg \max_{T \subseteq S} R_{\text{val}}^2(T), \quad (1)$$

where the subset is chosen on validation and the value is read on test. We write V_y for the closure throughout. It is the *ground truth* of this paper: every conclusion about risk is stated in terms of V_y computed by dedicated retraining.

3.3. Audit questions

The auditor asks three questions: (Q1) if a set S is published, how accurately can Y_y be inferred? (Q2) how risky is a single field i , why, and together with which other fields? (Q3) how can Q1–Q2 be answered for all fields and targets, repeatedly as the public baseline or target definitions change, without training a dedicated attacker for every set? Section 4 answers Q2 in terms of V_y , and Section 5 answers Q3.

4. Field-Level Inference Risk

4.1. Contextual gain

The quantity from which everything else follows is the gain that releasing field i brings over a given background T of fields the adversary already holds,

$$\Delta_i(A) = V_y(A \cup \{i\}) - V_y(A), \quad A = B \cup T. \quad (2)$$

$\Delta_i(B)$ is the single-field risk. The same field can be nearly useless over one background and decisive over another: a price series says little about a unit’s output alone, but together with the unit’s forecast availability it reveals the residual quantity the unit covers. A field-level risk quantity must therefore summarize a whole *distribution* of contextual gains $\{\Delta_i(B \cup T)\}_T$, and the question is which summary serves security.

4.2. Budgeted marginal risk and background amplification

Definition 3 (Budgeted marginal risk, background amplification). *For target y , public baseline B , fields under review $H = G \setminus B$ and budget K ,*

$$M_{i \rightarrow y}^{(K)} = \max_{T \subseteq H \setminus \{i\}, |T| \leq K} \Delta_i(B \cup T), \quad \Gamma_{i \rightarrow y}^{(K)} = M_{i \rightarrow y}^{(K)} - M_{i \rightarrow y}^{(0)} \geq 0. \quad (3)$$

Any maximizer T^ is a witness background.*

$M_{i \rightarrow y}^{(K)}$ is the largest increase of inferability that releasing i can cause when the adversary holds at most K other fields; $\Gamma_{i \rightarrow y}^{(K)}$ is the part of it that exists only because of the background, i.e. the amplification of single-field risk by combination. For each (field, target) pair the audit reports the profile $(M_{i \rightarrow y}^{(0)}, \Gamma_{i \rightarrow y}^{(K)}, M_{i \rightarrow y}^{(K)})$ together with the top- r backgrounds ranked by Δ_i ; in our data the top backgrounds are often nearly tied and share a core partner field, so the partner is more informative than a single argmax. At $K = 1$ the amplification has a closed form.

Proposition 1 (Amplification at $K = 1$). *With $I_{ij}^B = V_y(B \cup \{i, j\}) - V_y(B \cup \{i\}) - V_y(B \cup \{j\}) + V_y(B)$, $\Gamma_{i \rightarrow y}^{(1)} = \max(0, \max_j I_{ij}^B)$.*

Proof. Over the background $\{j\}$, $\Delta_i(B \cup \{j\}) = M_{i \rightarrow y}^{(0)} + I_{ij}^B$; over $\emptyset$ it is $M_{i \rightarrow y}^{(0)}$. Take the maximum. $\square$

A field is amplified by one background field exactly when it is strictly super-additive with it, and the amplification equals the strongest positive interaction.

4.3. Why the maximum

Turning a set function into a per-field score is the problem of feature attribution, and the natural first choice is an average of Δ_i over backgrounds, as in Shapley values or SAGE [17]. The average answers “how much does i contribute typically”, which is the right question for explanation and fair allocation. A security audit asks instead whether *some* admissible background makes i dangerous; a field whose gain is small over most backgrounds but large over one is dangerous, and averaging hides it—in our data the average contextual gain is typically about a third of the maximum (Section 8). This is the same reason quantitative information flow measures leakage by worst-case vulnerability rather than by average entropy [8, 9]. The maximum

Table 1: Requirements for a field-level inference-risk quantity. $\checkmark$: satisfied; $\circ$: partially; $-$: not satisfied.

Method	checkable by attack	joint inference	worst case	no dilution	bounded K	public baseline B
Correlation / mutual information	-	-	-	$\checkmark$	-	-
Single-field retraining $M^{(0)}$	$\checkmark$	-	-	$\checkmark$	-	$\circ$
Inference-graph propagation [6]	-	$\circ$	-	$\circ$	-	-
Shapley / SAGE [17]	$\checkmark$	$\checkmark$	-	-	-	-
LOCO / permutation	$\checkmark$	$\checkmark$	-	-	-	-
MCI [7]	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	-	-
$M_{i \rightarrow y}^{(K)}$ (this paper)	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$

marginal contribution is the Marginal Contribution Index (MCI) [7]; $M_{i \rightarrow y}^{(K)}$ adds the two conditions a disclosure audit needs, a public baseline B and a bounded background K . Table 1 summarizes the requirements and which field scores meet them.

Proposition 2 (Relation to MCI). *Let $v_B(T) := V_y(B \cup T) - V_y(B)$ for $T \subseteq H$. Then v_B is monotone, non-negative and $v_B(\emptyset) = 0$; $M_{i \rightarrow y}^{(K)} = \max_{|T| \leq K} [v_B(T \cup \{i\}) - v_B(T)]$ is the budget-constrained MCI of v_B ; for $K \geq |H| - 1$ and $B = \emptyset$ it is the MCI of V_y .*

The axiomatic properties of MCI for monotone evaluation functions [7]—dummy, symmetry, the self-contribution lower bound $M_{i \rightarrow y}^{(K)} \geq M_{i \rightarrow y}^{(0)}$ and invariance to duplicated features—therefore transfer and are cited rather than claimed; $M_{i \rightarrow y}^{(K)}$ is non-decreasing in K and reaches the MCI at $K = |H| - 1$.

4.4. Security meaning

Theorem 1 (Threshold-free safety margin). *Fix i, y, B, K . A number $m \geq 0$ satisfies “for every threshold τ and every background $|T| \leq K$: $V_y(B \cup T) \leq \tau - m \Rightarrow V_y(B \cup T \cup \{i\}) \leq \tau$ ” if and only if $m \geq M_{i \rightarrow y}^{(K)}$. For a fixed τ , the smallest m with this property can be strictly smaller than $M_{i \rightarrow y}^{(K)}$.*

Proof. If $m \geq M_{i \rightarrow y}^{(K)}$ then $V_y(B \cup T \cup \{i\}) = V_y(B \cup T) + \Delta_i(B \cup T) \leq (\tau - m) + m$. Conversely, for any T choose $\tau = V_y(B \cup T) + m$; the property gives $\Delta_i(B \cup T) \leq m$. Strictness: with $B = \emptyset$, $K = 1$, $V_y(\{i\}) = 0.1$, $V_y(\{j\}) = 0.5$, $V_y(\{i, j\}) = 0.9$ one has $M_{i \rightarrow y}^{(1)} = 0.4$, yet at $\tau = 0.95$ the necessary margin is 0. $\square$

Thus $M_{i \rightarrow y}^{(K)}$ is the answer to “by how much can publishing i raise the adversary’s accuracy, whatever it already holds within budget”—a property of the field that does not depend on the threshold later chosen for grading.

Definition 4 (Criticality and minimal unsafe sets). *Under baseline B , $E \subseteq H$ is a τ -minimal unsafe set (MUS) if $V_y(B \cup E) > \tau$ and $V_y(B \cup E') \leq \tau$ for every $E' \subsetneq E$. Field i is K -background τ -critical if some T with $|T| \leq K$, $i \notin T$, has $V_y(B \cup T) \leq \tau < V_y(B \cup T \cup \{i\})$.*

Theorem 2 (Criticality $\iff$ small MUS). *If V_y is monotone and $V_y(B) \leq \tau$, field i is K -background τ -critical if and only if it belongs to a MUS of size at most $K + 1$.*

Proof. ($\Leftarrow$) For a MUS $E \ni i$ with $|E| \leq K + 1$ take $T = E \setminus \{i\}$. ($\Rightarrow$) If $V_y(B \cup T) \leq \tau < V_y(B \cup T \cup \{i\})$, take an inclusion-minimal unsafe $E \subseteq T \cup \{i\}$; it is a MUS of size $\leq K + 1$. If $i \notin E$ then $E \subseteq T$ and monotonicity gives $V_y(B \cup T) \geq V_y(B \cup E) > \tau$, a contradiction. $\square$

Theorem 2 makes the field-level flag a faithful projection of the set-level danger structure: flagging the τ -critical fields misses no field that takes part in a dangerous combination of at most $K + 1$ fields, and the witness T^* names the combination. The same list of minimal unsafe sets also supports downstream disposal (Section 12). Monotonicity is essential: without the closure, a field can cross the threshold over a background that already contains an unsafe subset, and no MUS then contains it.

4.5. Choice of K

Different K describe adversaries with different background knowledge, and $M_{i \rightarrow y}^{(K)}$ grows with K . We report $K = 2$ as the main setting and measure $K = 3$ on every data set. The threat model is an adversary holding a handful of other published series rather than the whole catalogue; high-order interactions are rare in engineered systems [37], and models with pairwise interactions are close to unrestricted ones on tabular data [38]. There is, however, no general saturation theorem—a parity function of $K + 2$ fields defeats any fixed K —and saturation of the maximal gain does not imply that no larger set crosses a threshold. Section 11 measures both.

5. An Amortized Attacker for Set-Level Inferability

5.1. Why an estimator is needed

Evaluating $M_{i \rightarrow y}^{(K)}$ for all fields at $K = 2$ requires V_y on every set of size at most three; with 35 candidates this is 7,175 sets per target, each requiring a family of attackers to be trained. The table has to be recomputed whenever

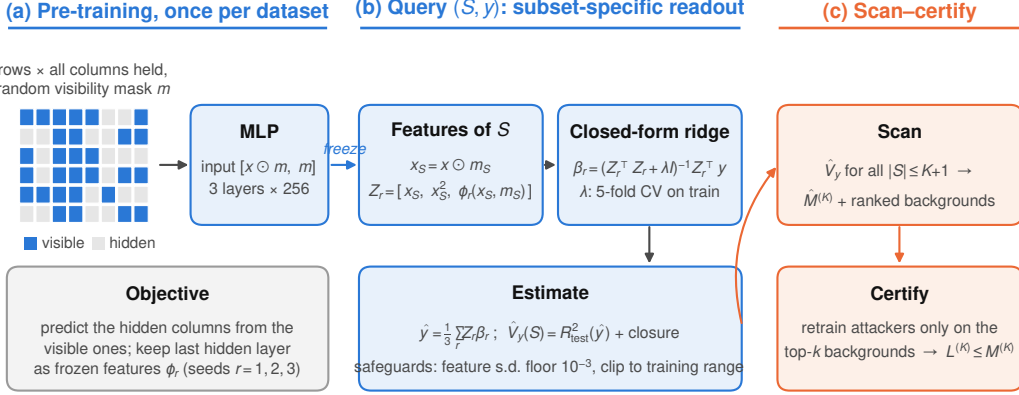


Figure 2: The amortized attacker. (a) One backbone is pre-trained under random visibility masks on the auditor’s data and frozen. (b) A query (S, y) solves a ridge readout on the features of S in closed form, per seed, and averages the predictions. (c) The scan proposes backgrounds per field; retraining only these certifies a lower bound on $M_{i \rightarrow y}^{(K)}$.

the public baseline grows, a target is redefined, or a what-if release is examined. We therefore build one model that answers $V_y(S)$ for any S quickly (Fig. 2), and we make sure that no conclusion depends on its accuracy alone (Section 5.6).

5.2. Masked pre-training

The adversary does not know in advance which fields it will obtain. We model this by pre-training one network under random visibility masks. A mask $m \in \{0, 1\}^p$ is drawn by first sampling the number of visible fields uniformly and then a uniform subset of that size; the network $f_\theta(x \odot m, m)$ receives the masked values and the mask itself, so that a hidden field is distinguishable from a zero value. The objective combines predicting the sensitive targets from the visible fields with, optionally, reconstructing every hidden column from the visible ones [30, 31],

$$\mathcal{L}(\theta) = \mathbb{E}_m \left[\|f_\theta^y(x \odot m, m) - y\|^2 + \gamma \frac{\|(1 - m) \odot (f_\theta^x(x \odot m, m) - x)\|^2}{\|1 - m\|_1} \right]. \quad (4)$$

Target-only pre-training ($\gamma = 0$) is the estimator E used for the main results. We found that it collapses the hidden layer onto the few directions that predict the targets on average over masks, and that on the small sets of an audit it then adds nothing over an untrained network; reconstruction

($\gamma = 1$) forces the representation to encode the operating state that drives all fields and improves every accuracy metric (Section 10). Reconstructing all columns the auditor holds also decouples the backbone from any particular target definition. The last hidden layer (256 units) is kept as a frozen feature map $\phi_\theta(x \odot m, m)$; early stopping uses the target loss on validation rows only.

5.3. Subset-specific closed-form readout

For a query (S, y) with mask m_S and seed r , the feature block is

$$Z_r(S) = [x \odot m_S, (x \odot m_S)^2, \phi_{\theta_r}(x \odot m_S, m_S)], \quad (5)$$

standardized with training statistics. A ridge readout

$$\beta_r(S) = (Z_r^\top Z_r + \lambda I)^{-1} Z_r^\top y \quad (6)$$

is solved in closed form on the training rows, with λ chosen by five-fold cross-validation inside the training rows. The predictions of $R = 3$ independently pre-trained backbones are averaged, $\hat{y} = R^{-1} \sum_r Z_r^{\text{test}} \beta_r(S)$, and $\hat{V}_y(S) = R_{\text{test}}^2(\hat{y})$, followed by the monotone closure. Numerical safeguards matter on real data: a floor of 10^{-3} on feature standard deviations, and clipping test features to their training range. Without them, units that are almost constant on the training rows (inactive ReLUs) are amplified on test rows, a few predictions explode and $\hat{V}$ collapses to zero for those sets; since $M_{i \rightarrow y}^{(K)}$ is a maximum over differences, such collapses turn directly into inflated risk (Section 10). A reconstruction backbone needs a third safeguard, clipping predictions to the range of the target on the training rows. No test label enters pre-training, the readout, the choice of λ or the closure.

5.4. Why the readout is solved per subset

A cheaper alternative lets the network output the prediction directly, $\hat{y} = f_\theta(x \odot m_S, m_S)$, and reads $\hat{V}$ off it. Then one set of output weights serves every conditioning pattern, i.e. the attacker is forced to combine every subset of fields in the same way—the very effect that combination risk is about. Solving the readout per subset keeps the expensive representation shared and the combination rule specific.

5.5. Error propagation

Write the estimate as $\hat{V}(S) = V(S) + b + \varepsilon(S)$, where b is a bias common to the sets of interest and $\varepsilon(S)$ the set-specific part. Since a marginal is a difference of two nested sets, $\hat{\Delta}_i(T) - \Delta_i(T) = \varepsilon(T \cup \{i\}) - \varepsilon(T)$ and

$$|\hat{M}_{i \rightarrow y}^{(K)} - M_{i \rightarrow y}^{(K)}| \leq \max_{|T| \leq K} |\varepsilon(T \cup \{i\}) - \varepsilon(T)|, \quad (7)$$

independently of b . A readout on a frozen representation typically underestimates the fully trained attacker by a similar amount on nested sets, which is exactly a common-mode b ; this is why an estimator can be biased in V yet accurate in M . The quantity to control is the *marginal* error, which we report throughout.

5.6. Scan and certify

The estimator is used to *scan*: it produces $\hat{M}^{(K)}$ for every field together with a ranked list of candidate backgrounds. Decisions are then *certified*: for each field, dedicated attackers are retrained on the top- k backgrounds only, and $L_{i \rightarrow y}^{(K)} = \max_{T \in \text{top-}k} \Delta_i(B \cup T) \leq M_{i \rightarrow y}^{(K)}$ is a lower bound computed entirely from retrained attackers. The ratio $\sum_i L / \sum_i M$ measures how much of the true risk the scan recovers at a certification cost of k retrained backgrounds per field instead of all $\binom{|H|-1}{\leq K}$.

6. Experimental Setup

6.1. Data sets and rule-based field roles

Table 2 summarizes the four data sets. In all of them, candidate fields are information that the disclosure rules [1] release *before* the target, and targets are either specific information (unit output, Art. 17(3)), quantities not on the disclosure list at hourly resolution (hourly line flow; only daily averages of important lines are listed), or actual operating quantities released on the following day (actual load, total generation, tie-line exchange; Appendix items 6.57–6.66). Using the release timing to assign roles does not involve any temporal split of the data; all splits are random.

RTS-GMLC. The IEEE Reliability Test System 2019 update [39] provides a 73-bus network with one year of realistic hourly load, wind, solar and hydro series. We clear an hourly DC optimal power flow with piecewise-linear offers built from the published heat-rate curves (8,784 hours, 26.7%

Table 2: Data sets. n : train/test rows; p : candidate fields; sets: field sets with exact three-attacker ground truth.

Data set	Targets (rule basis)	n	p	sets	max size
RTS-GMLC	unit 223, unit 221 (specific info.); line C35 (not listed)	6149/2635	23	10,902	4
NEM 2024	units BW01, TUMUT3, PPCCGT (specific info.)	6149/2635	30	31,930	4
PJM (load)	metered load (next-day)	4589/1967	22	9,108	4
PJM (gen./interch.)	total generation, net interchange (next-day)	4589/1967	24	12,950	4
CAISO	actual load (next-day)	6105/2617	35	59,535	4

congested). Candidates are the 23 fields a market would publish early (zonal load forecasts, renewable and hydro forecasts, reserve requirements, system energy price and nine nodal prices, three congestion indicators); targets are the outputs of a coal and a combined-cycle unit and the hourly flow on internal line C35.

NEM. One year (2024) of the Australian National Electricity Market [40], averaged to hours. Candidates are regional demand, available capacity, semi-scheduled wind and solar forecasts, regional prices and interconnector flows (30 fields); targets are the actual outputs of a coal unit (BW01), a hydro unit (TUMUT3) and a gas combined-cycle unit (PPCCGT). Unit output is public in the NEM but is specific information under the Chinese rules; this data set therefore contains a genuinely sensitive field observed in the real world.

PJM and CAISO. System-level hourly data for one year. Targets are the actual quantities released on the next day: PJM metered load, total generation and net actual interchange, and CAISO actual load. Candidates are forecasts, day-ahead and real-time prices and their components, ancillary-service requirements and cleared volumes, and schedules (22–35 fields). Fields released on the same day as the target (fuel-mix generation, actual renewable output, other balancing areas’ actual load) are excluded, as are the load forecasts of the target area, which are ex-ante copies of the target (Section 3). Near-copies that are distinct quantities (e.g. sub-area forecasts for CAISO load) are kept; removing them is analysed as a sensitivity case.

Controlled mechanism. A fifth data set is built on the RTS-GMLC market with a private hourly offer markup of one unit as the target and eleven fields chosen to span the known mechanism (Section 7); all 2,047 of its field sets are retrained.

6.2. Ground truth

All variables are standardized with training statistics; 70/30 random train/test splits are used, with 15% of the training rows as validation. The attacker family has three members: a single-target multilayer perceptron (two hidden layers of 128 units, dropout 0.15, Adam, early stopping on validation), a multi-target perceptron trained on all targets of a data set, and histogram gradient boosting with two configurations. For every set of up to four fields, every member is trained on that set alone; the member and epoch are chosen on validation and the test R^2 is reported, followed by the monotone closure (1). This amounts to 124,425 field sets and several hundred thousand trained models. Neither the estimator nor any baseline has access to these values; they are used only for evaluation.

6.3. Baselines

Field scores compared with $M^{(2)}$ are: Pearson r^2 , squared Spearman correlation, mutual information [41], distance correlation [42], single-field re-training $M^{(0)}$ (current practice), LOCO [20], permutation importance [21], SAGE with marginal imputation [17], and an inference-graph score that follows the propagation idea of [6] without its expert inputs: edge weights are the single-field predictability of one field from another (gradient boosting, validation R^2), the target node carries risk one, and risk is propagated over three hops with decay 0.5. All baseline scores are computed on training rows only.

Estimators compared with ours are: linear and quadratic readouts without a backbone, an in-set quadratic dictionary (values, squares and pairwise products of the fields in S), the shared output head of the same backbone, an untrained (randomly initialized) backbone with the same readout, and warm-start fine-tuning of the whole network for each set [25]. Pre-training objectives are compared on the same sets with three further backbones per data set: masks restricted to small visible sets, masked reconstruction of all candidate columns, and both.

6.4. Metrics

For set values: mean absolute error of $\hat{V}$. For field risk: mean absolute error and bias of $\hat{M}^{(2)}$, Spearman correlation across fields, and the certified ratio $\sum_i L_{i \rightarrow y}^{(2)} / \sum_i M_{i \rightarrow y}^{(2)}$ when k backgrounds per field are retrained. For decisions at thresholds $\tau \in \{0.5, 0.7\}$: recall and precision of critical fields;

the number of fields withheld and the number of minimal unsafe sets of size ≤ 3 left unbroken.

7. Results I: A Controlled Power-System Mechanism

Before turning to real markets we test the definitions on a case whose inference mechanism is known in advance. We take the RTS-GMLC market of Section 6.1 and give one generating unit, 313_CC_1 (a 355 MW combined-cycle unit that is partly loaded in about 90% of hours and often sets the price), a private hourly offer markup: all its offer segments are multiplied by $(1 + \kappa_t)$, with κ_t drawn independently each hour. Offers are private information under the disclosure rules, and κ_t is the sensitive target. We clear all 8,784 hours again and assemble eleven fields: the unit’s output P_g , the LMP at its bus, the LMP at a nearby bus of the same area, the LMP at a bus behind the most frequently congested line (C6), an LMP in another area, system and area load, available wind and solar, the congestion indicator of C6, and the output of a competing unit. With eleven fields every one of the 2,047 subsets is retrained, so $M_{i \rightarrow y}^{(K)}$ is exact for every K up to the full MCI.

Prediction. When the unit is marginal, its bus price equals $(1 + \kappa_t) c_s$, where c_s is the cost of the segment in which it operates, and P_g identifies that segment. Hence $\{P_g, \lambda_g\}$ reveals κ_t while neither field does so alone; a nearby price should substitute for λ_g when the network is uncongested, a price behind a congested line should not, and loads, renewables and the congestion flag can only indicate whether the unit is marginal. We use small markups, $\kappa_t \sim U(-5\%, 5\%)$, which leave the merit order almost unchanged; larger markups ($\pm 30\%$) change the dispatch itself and are reported as a contrast.

Results. Table 3 and Fig. 3 confirm each prediction. Every price field has zero single-field risk and a budgeted risk of 0.10–0.15: its risk is *entirely* background amplification, and the witness is always the unit’s output. The gain of P_g itself is 0.065 alone and ranks its backgrounds exactly as the mechanism implies—own-bus price (0.215), nearby price (0.191), other-area price (0.172), remote price (0.163), competitor output (0.151), and loads, renewables and congestion (≈ 0.07). The risk also depends on the operating regime: in hours when C6 is congested the remote price loses its value as a background (gradient-boosting attacker; the joint inferability with P_g falls from 0.185 to 0.012), whereas the own-bus price becomes more informative

Table 3: Controlled mechanism (markup $\pm 5\%$): exact field risk over all 2,047 subsets.

Field	$M^{(0)}$	$M^{(2)}$	MCI	witness at $K = 2$
Unit output P_g	0.065	0.215	0.215	own-bus LMP
Own-bus LMP	0.000	0.150	0.150	P_g
Nearby LMP (306)	0.000	0.141	0.141	$P_g + C6$
Other-area LMP (223)	0.000	0.115	0.115	$P_g + C6$
Remote LMP (303, behind C6)	0.000	0.099	0.099	P_g
Competitor output (316)	0.000	0.086	0.086	P_g
Loads, renewables, C6	0.000	0.02–0.05	≤ 0.095	–

(0.198 to 0.289). Because the mechanism is pairwise, $M^{(1)} = M^{(2)} = \dots =$ MCI for every price field. With $\pm 30\%$ markups the output alone already reveals much of the markup ($M^{(0)} = 0.52$), and the prices remain amplified from zero to about 0.14: the smaller the markup, the more completely it hides from single-field screening, but not from the combination.

The scores of Section 6 behave as Table 1 predicts (Fig. 3c). Correlation, single-field retraining and pairwise inference-graph propagation assign about zero to every price. SAGE assigns 0.003 to the own-bus price, because its credit is shared among four mutually substitutable prices; LOCO likewise sees almost nothing once another price is present, and permutation importance recognizes only the one price the fitted model happens to use. $M^{(2)}$ flags all four prices and the competitor output and names the unit output as their background.

8. Results II: Background Amplification in Real Data

Single-field grading misses most of the risk. We first ask how much of the risk in the four data sets is invisible to single-field grading. Table 4 and Fig. 4 report, for each of the ten targets, how many candidate fields are unsafe on their own and how many become critical only in combination.

Of the 264 field–target pairs, 256 are critical at $\tau = 0.5$ with at most two background fields, and 223 of these are safe on their own; at $\tau = 0.7$ the figures are 138 and 113. For four targets (NEM BW01 and TUMUT3, PJM load and PJM net interchange) no candidate exceeds $\tau = 0.5$ alone, yet 22 to 30 candidates take part in a dangerous combination; at $\tau = 0.7$ the same holds for RTS line C35, where 21 of 23 fields are critical and none is unsafe alone. Withholding every field that is unsafe alone breaks almost none of

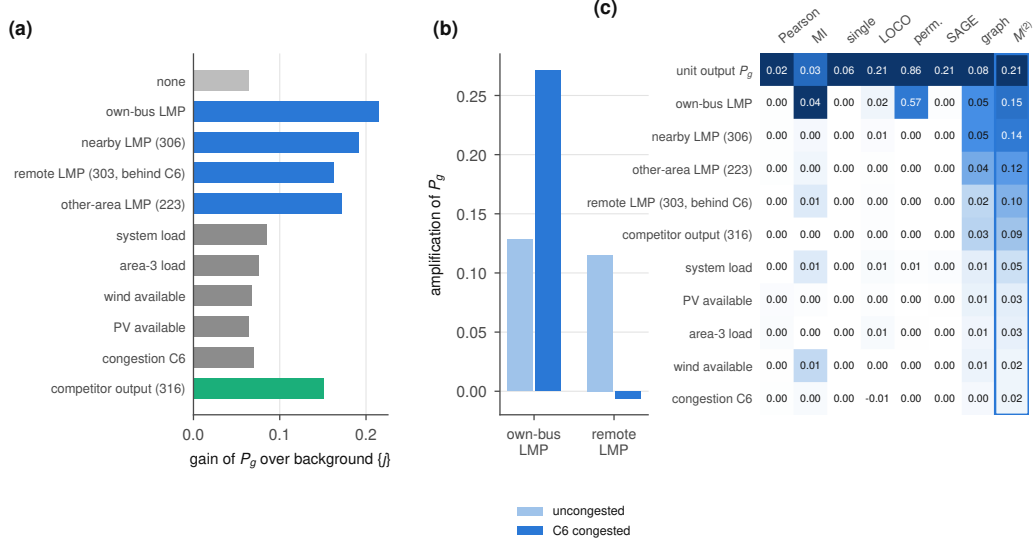


Figure 3: Controlled mechanism: private offer markup of unit 313_CC_1 ($\pm 5\%$). (a) Gain of the unit output P_g over each single background. (b) Amplification of P_g by the own-bus and the remote price in uncongested hours and in hours when line C6 is congested. (c) Scores of eight methods for the eleven fields (colour normalized per column; last column $M^{(2)}$).

Table 4: Combination risk that single-field grading misses. V_1 : best single field; $V_{\leq 3}$: best set of at most three fields; alone: fields with $V(\{i\}) > \tau$; comb.: fields safe alone but τ -critical with at most two background fields; MUS: minimal unsafe sets of size ≤ 3 ; left: MUS not broken after withholding every field that is unsafe alone; hit: minimum number of fields whose withholding breaks every MUS.

Target	p	V_1	$V_{\leq 3}$	$\tau = 0.5$					$\tau = 0.7$				
				alone	comb.	MUS	left	hit	alone	comb.	MUS	left	hit
RTS unit 223	23	1.00	1.00	10	13	60	50	15	10	6	16	6	11
RTS unit 221	23	0.99	0.99	10	13	52	42	15	10	9	34	24	13
RTS line C35	23	0.52	0.86	4	19	123	119	13	0	21	82	82	7
NEM BW01	30	0.49	0.65	0	30	266	266	4	0	0	0	0	0
NEM TUMUT3	30	0.38	0.53	0	22	23	23	1	0	0	0	0	0
NEM PPCCGT	30	0.55	0.80	2	28	215	213	6	0	10	8	8	1
PJM load	22	0.48	0.80	0	22	237	237	9	0	14	44	44	3
PJM total gen.	24	0.97	0.99	2	22	179	177	11	2	18	39	37	5
PJM net interch.	24	0.41	0.71	0	24	102	102	3	0	3	1	1	1
CAISO load	35	0.82	0.93	5	30	542	537	17	3	32	335	332	10

the minimal unsafe sets (column “left”): 1,766 of 1,799 at $\tau = 0.5$ and 534 of 559 at $\tau = 0.7$ remain.

Where single fields do leak, combinations still dominate. Some targets are exposed by a single close proxy: RTS unit outputs are almost determined by a nodal price ($V_1 = 0.99$), and PJM total generation by the system load forecast (0.97). For these, $M^{(0)}$ already flags the proxy, and the remaining risk is again combinatorial (e.g. 22 of 24 PJM fields are combination-critical). Removing such near-proxies from the candidate pool (Section 6.1) lowers the best single field to 0.42 for both PJM total generation and CAISO load but leaves every remaining field combination-critical at $\tau = 0.5$ (22/22 and 26/26), so the conclusion does not depend on where the proxy boundary is drawn.

The witnesses are physically meaningful. Fig. 5 shows the field profile of RTS line C35 and the gain matrix of NEM unit PPCCGT. The largest escalations come from combinations that reconstruct a residual quantity: the SP15 day-ahead energy price rises from $M^{(0)} = 0.14$ to $M^{(2)} = 0.51$ for CAISO load with the NP15 wind and SP15 solar forecasts as witnesses (price and renewables determine net load); the South Australian wind forecast rises from 0.19 to 0.60 for PPCCGT together with the regional available capacity and solar forecast (the residual demand the gas unit is dispatched for); the price at RTS bus 303 rises from 0.36 to 0.65 for the C35 flow together with a zonal load forecast and the congestion indicator of the neighbouring line C6. The gain matrix also shows that the same field is harmless next to some partners and decisive next to others, which is what a grade that ignores the background cannot represent.

Contextual gains are widely spread. Fig. 6(b) plots, for the most amplified field of six targets, the gain $\Delta_i(T)$ over every background of at most two fields. The spread is large, and the maximum typically lies far from the bulk: the South Australian wind forecast has gain 0.19 alone, 0.20 on average over backgrounds, and 0.60 over its worst background (regional available capacity and solar forecast, which together with wind give the residual demand the gas unit PPCCGT covers). Over all 264 field–target pairs, the median ratio of the average contextual gain to the maximum is 0.32: an average-based score would report about a third of the worst-case risk.

Amplification is common. Averaged over the 264 pairs, $\Gamma^{(1)} = 0.082$, $\Gamma^{(2)} = 0.110$ and $\Gamma^{(3)} = 0.119$; one background field supplies 75% of the total amplification at $K = 2$, and $K = 2$ supplies 93% of that at $K = 3$. In

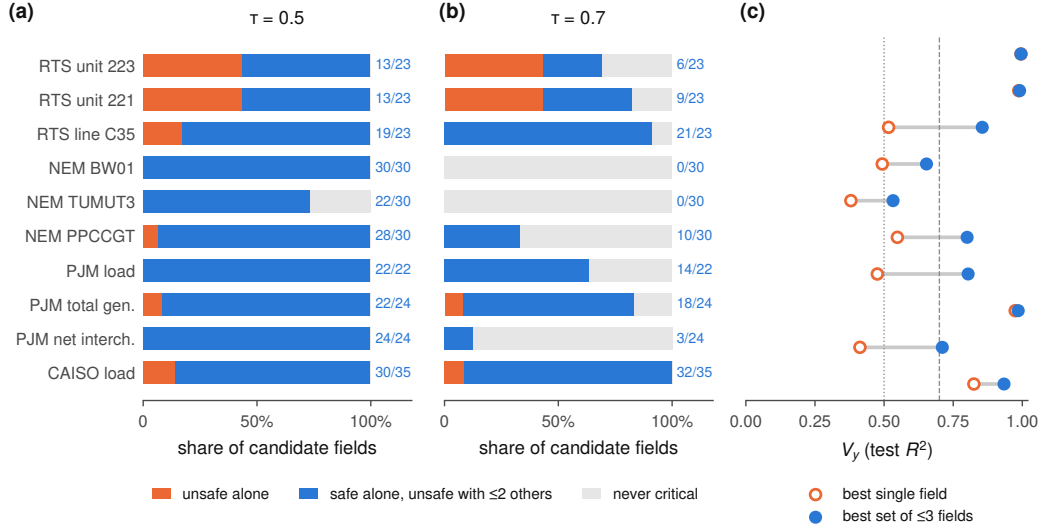


Figure 4: Share of candidate fields that are unsafe alone (orange), safe alone but critical with at most two background fields (blue), or never critical (grey), at (a) $\tau = 0.5$ and (b) $\tau = 0.7$; numbers give combination-critical fields over all candidates. (c) Best single field versus best set of at most three fields.

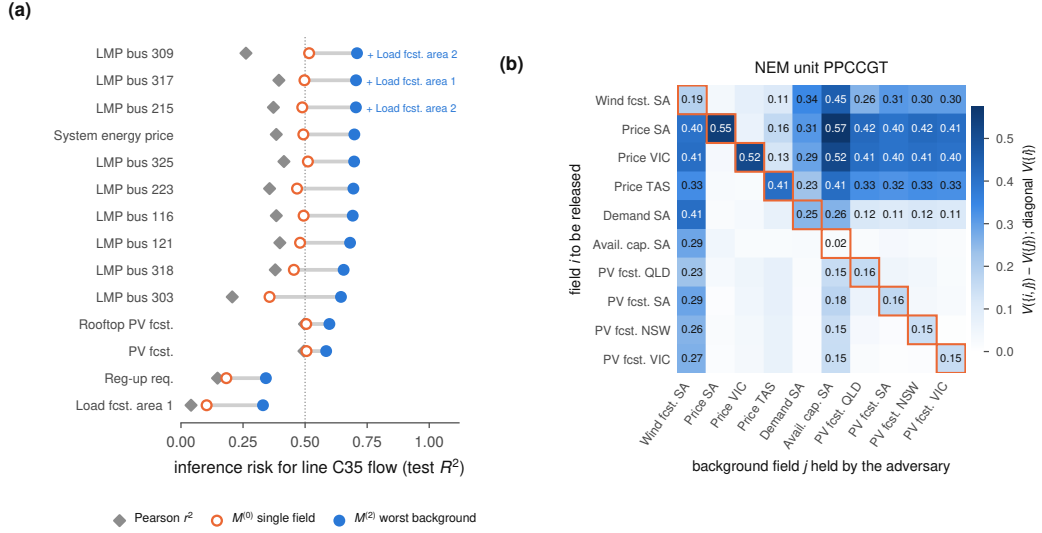


Figure 5: (a) Field profile for the hourly flow on RTS line C35 (14 fields with the largest $M^{(2)}$): Pearson r^2 , single-field risk and worst-background risk, with the witness background for the three riskiest fields. (b) Gain matrix for NEM unit PPCCGT: entry (i, j) is $V(\{i, j\}) - V(\{j\})$, the diagonal is $V(\{i\})$; values below 0.10 are not printed.

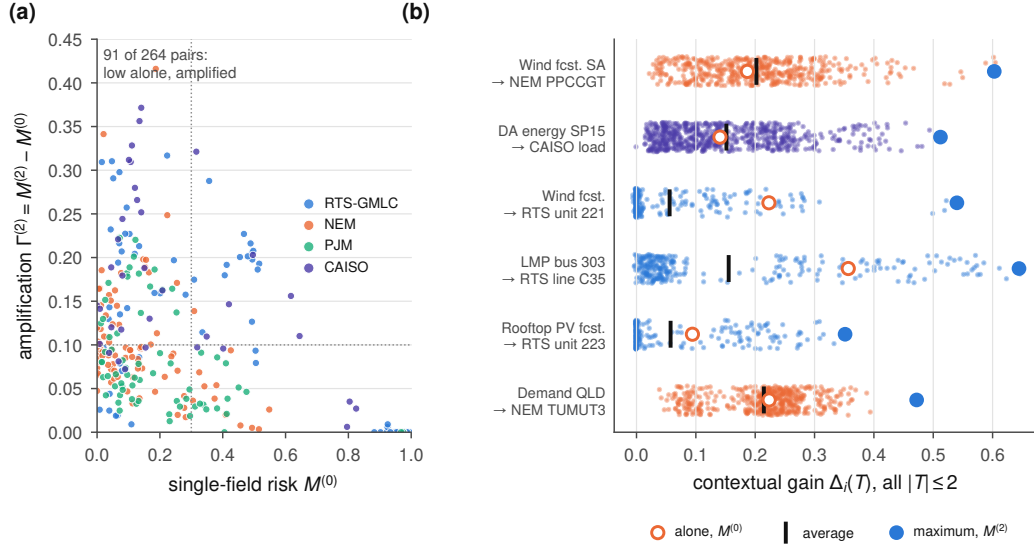


Figure 6: Background amplification over ten targets. (a) Single-field risk versus amplification $\Gamma^{(2)}$ for every field–target pair. (b) Contextual gains of the most amplified field of six targets over all backgrounds of at most two fields, with the single-field value, the average over backgrounds and the maximum.

91 pairs the single-field risk is below 0.3 while the amplification exceeds 0.1 (Fig. 6a)—fields that look unremarkable alone and become informative in combination. The largest amplifications are the SA wind forecast for PPCCGT (0.19 → 0.60), the SP15 day-ahead energy price for CAISO load (0.14 → 0.51, with the NP15 wind and SP15 solar forecasts: price and renewables determine net load), and the SA available capacity for PPCCGT (0.02 → 0.36). On every pair, $\Gamma^{(1)}$ equals the strongest positive pairwise interaction (Proposition 1) to numerical precision.

Dangerous backgrounds come in families. The top backgrounds of a field are often nearly tied and share a core partner. For the SA wind forecast the five best backgrounds reach gains between 0.545 and 0.603, and all contain the SA available capacity. We therefore report the top- r backgrounds rather than a single witness; the shared partner is the actionable information (“do not co-release with”).

9. Results III: Comparison of Field-Level Risk Scores

The mechanism case (Section 7) already shows how each family of scores fails on a known structure; a purely synthetic example with the same three failure modes—synergy, redundancy and indirect contribution—is given in Appendix C. Here we compare the scores as rankings of the fields of the ten real targets, against critical fields defined by retrained ground truth.

Ranking. Table 5 compares the scores as rankings. At $\tau = 0.7$, $\hat{M}^{(2)}$ has the highest R-precision (0.903, against 0.866 for mutual information, 0.830 for the inference graph, 0.803 for single-field retraining, 0.769 for SAGE and 0.727 for LOCO), and withholding fields in the order of exact or certified $M^{(2)}$ exceeds the optimum by only 0.4 fields. At $\tau = 0.5$ almost every field is critical and every score reaches an R-precision of about 0.98; here single-field retraining ranks as well as $M^{(2)}$. We report this plainly: as a *ranking*, single-field risk is a strong baseline when most fields are dangerous; the advantage of $M_{i \rightarrow y}^{(K)}$ lies in decisions, disposal and high thresholds. Attribution scores computed on one fitted model (LOCO, permutation importance, SAGE) are the weakest throughout, because they measure what that model uses rather than what an adversary can infer.

10. Results IV: The Amortized Attacker

Fidelity. Fig. 7 compares the amortized estimates with retrained ground truth for all sets of size ≤ 3 . The estimate of V is biased downward, visibly so for PJM, where the training set is smallest; averaged over the ten targets the error is 0.026 with a signed bias of -0.024 , i.e. almost entirely common-mode. As predicted by (7), the field-level risk is estimated more accurately than the set values it is built from: the mean absolute error of $\hat{M}^{(2)}$ is 0.021 and its Spearman correlation with the exact $M^{(2)}$ across fields is 0.957 (Table 6). Certification recovers most of the exact risk from very few retrained backgrounds: retraining the single best-scanned background per field yields a lower bound equal to 93.4% of the exact $M^{(2)}$ on average, three backgrounds yield 97.0%, and on RTS-GMLC and CAISO the top-3 ratio exceeds 99% (Fig. 7c). For a field set of 35 candidates this replaces the $1 + 34 + \binom{34}{2} = 596$ backgrounds per field of an exhaustive evaluation by three.

Table 5: Ranking quality over the ten targets. R-precision: share of true critical fields among the top- c fields, c being the true number of critical fields (a threshold that favours the baselines). Excess: fields withheld in score order until no MUS remains, minus the optimum.

Score	R-precision		excess withheld	
	$\tau = 0.5$	$\tau = 0.7$	$\tau = 0.5$	$\tau = 0.7$
Pearson r^2	0.977	0.815	5.4	3.0
Spearman ρ^2	0.977	0.767	3.5	2.8
Mutual information	0.977	0.866	2.0	2.1
Distance correlation	0.982	0.833	3.0	2.0
Single-field $V(\{i\})$	0.982	0.803	1.6	1.9
LOCO	0.977	0.727	11.0	8.0
Permutation importance	0.968	0.795	8.9	4.7
SAGE	0.968	0.769	8.6	6.3
Inference-graph propagation	0.982	0.830	5.6	2.7
$\hat{M}^{(2)}$ (amortized, no retraining)	0.982	0.903	3.0	0.9
$M^{(2)}$ scan-certify ($k = 3$)	0.982	0.888	1.6	0.4
$M^{(2)}$ exact	0.977	0.894	1.5	0.4

Amplification and dangerous backgrounds. The profile of Section 8 is estimated with similar accuracy: the mean absolute error of $\hat{\Gamma}^{(2)}$ is 0.017. Identifying the single worst background is harder—the best-scanned background is the worst one for 47% of the fields (17–78% across targets), and the worst one is among the three best-scanned for 67% and among the five best-scanned for 74%—but this matters little, because dangerous backgrounds come in families of nearly tied members: retraining the three best-scanned backgrounds per field recovers 92.8% of the total amplification.

What matters in the estimator. Table 6 and Fig. 8(a–c) separate the components. The largest single effect is solving the readout per field set: with the same backbones, reading the prediction off a shared output head gives an $M^{(2)}$ error of 0.057, against 0.021 for the per-set readout, and a lower certified ratio. The second is a learned nonlinear representation: without a backbone, a linear readout (0.144) or an in-set quadratic dictionary (0.077) is far less accurate. Masking during pre-training is necessary—a backbone trained on fully visible inputs is worse than the per-set readout on any other backbone (0.036, Spearman 0.928)—but the masking distribution is not critical (Bernoulli versus uniform visible count: 0.024 versus 0.023). The numeri-

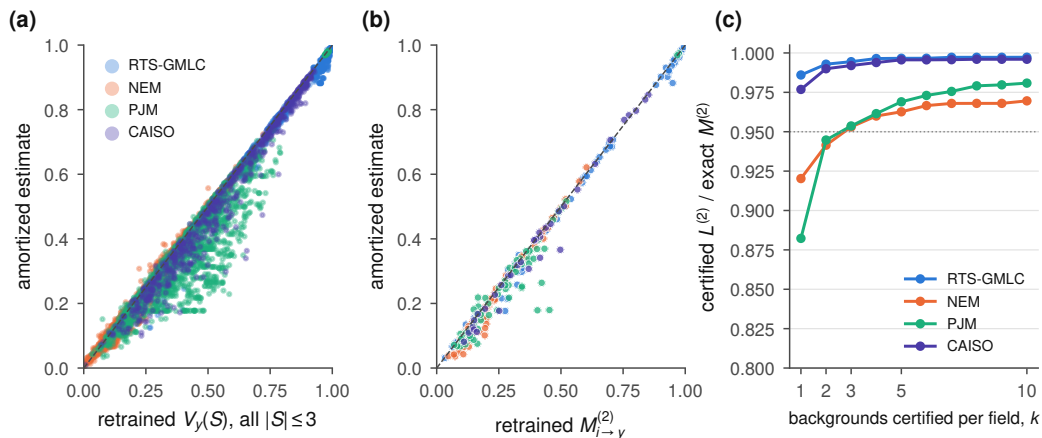


Figure 7: Amortized estimates against retrained ground truth. (a) $\hat{V}_y(S)$ for sets of size ≤ 3 (random subsample). (b) $\hat{M}^{(2)}$ for every field and target. (c) Certified lower bound over exact $M^{(2)}$ when the top- k scanned backgrounds per field are retrained.

Table 6: Estimator comparison and ablation, averaged over the ten targets and all sets of size ≤ 3 . Ablations use one seed and should be compared with “ours, one seed”. Cert.@ k : certified $L^{(2)}$ over exact $M^{(2)}$ when k backgrounds per field are retrained.

Estimator	MAE V	MAE Δ	MAE $M^{(2)}$	bias $M^{(2)}$	Spearman	cert.@1	cert.@3
Linear readout on $x \odot m$ (no backbone)	0.212	0.068	0.144	-0.139	0.737	0.803	0.878
Readout on $[x, x^2]$ (no backbone)	0.132	0.049	0.096	-0.089	0.854	0.859	0.912
In-set quadratic dictionary (no backbone)	0.120	0.045	0.077	-0.066	0.870	0.872	0.932
Shared output head (3 seeds)	0.067	0.030	0.057	-0.054	0.930	0.883	0.939
Untrained random backbone + readout	0.027	0.017	0.022	-0.015	0.959	0.906	0.971
Pre-training without masks	0.032	0.020	0.036	-0.004	0.928	0.898	0.963
Bernoulli(0.5) masks	0.030	0.019	0.024	-0.018	0.958	0.932	0.969
Readout on ϕ only	0.030	0.019	0.025	-0.016	0.950	0.927	0.964
No numerical safeguards	0.031	0.020	0.036	+0.009	0.904	0.825	0.946
Ours , one seed	0.030	0.019	0.023	-0.019	0.953	0.936	0.967
Ours , three seeds (main)	0.026	0.017	0.021	-0.018	0.957	0.934	0.970

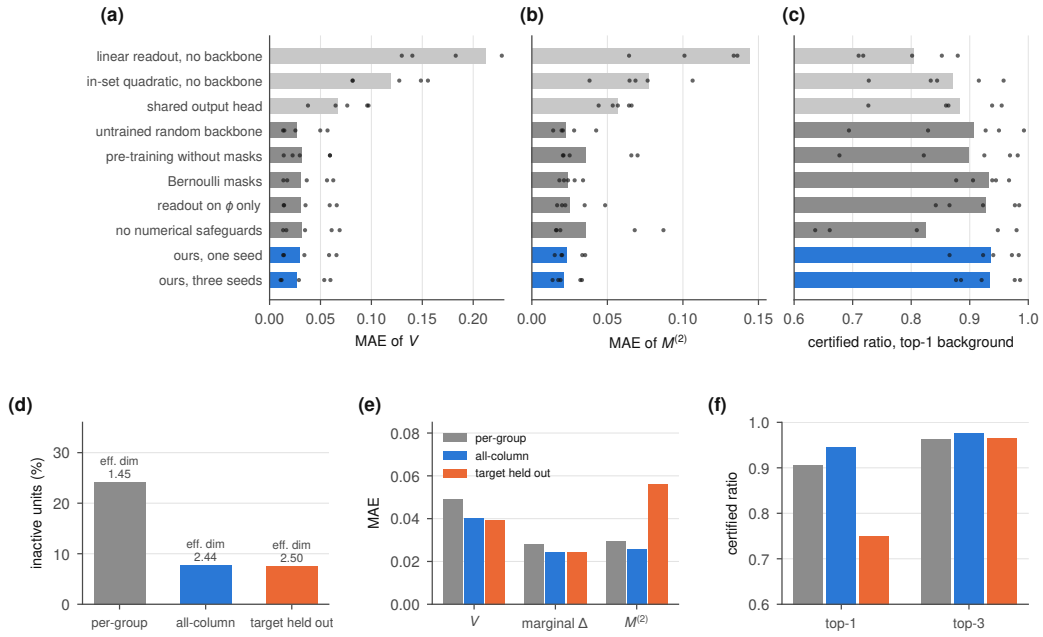


Figure 8: (a–c) Estimator comparison and ablation over ten targets (bars: mean; dots: data sets); light grey: no per-set readout or no backbone; dark grey: one-seed ablations; blue: ours. (d–f) Backbone scope on the four PJM/CAISO targets: inactive hidden units and effective dimension, error of V , of marginals and of $M^{(2)}$, and certified ratios.

cal safeguards remove the overestimation of $M^{(2)}$ on real data (bias +0.009 without them, certified top-1 ratio 0.825 instead of 0.936). Averaging three seeds reduces the V error from 0.030 to 0.026.

Why target-only masked pre-training adds little. One result of the ablation is counter-intuitive: an untrained, randomly initialized backbone with the same per-set readout estimates V and $M^{(2)}$ about as well as the backbones pre-trained on the targets (Table 7). Three findings explain it (Fig. 9). *(i) Collapse.* A backbone trained only to predict one to three targets compresses its hidden layer onto the few directions that predict them on average over masks: the effective dimension of the representation (participation ratio) is 1.60, against 3.30 for an untrained network. Restricting pre-training masks to one to four visible fields, the size of audit queries, does not change this (1.65), so the cause is the narrow objective, not the mask distribution. *(ii) What the readout uses.* Truncating the features of each set to their r leading principal components shows that pre-training does find target-aligned directions—at $r = 2$ its error is half that of untrained features—but that the readout draws most of its accuracy from the remaining directions, where random ReLU features serve as well (Fig. 9a). A ridge readout on 256 random features is a random-feature approximation of a kernel regression [43], which is already adequate for a function of at most three fields learned from several thousand rows. *(iii) Set size.* The gap opens for larger sets: for sets of 16 fields the error of untrained features is 0.074, against 0.027 for target-only and 0.023 for reconstruction pre-training (Fig. 9b; gradient-boosting ground truth). Audit queries are small, which is where pre-training matters least.

Repairing it. Adding the reconstruction term of (4) restores the effective dimension (2.75) and lowers the error of V by 24% (three seeds, sets of size ≤ 3 and of size four) and that of the marginals by 18%. At first, $M^{(2)}$ did not improve (0.027 against 0.021). Because $M_{i \rightarrow y}^{(K)}$ is a maximum over marginals, its error is governed by the upper tail of the marginal errors, not by their mean (Fig. 9c), and the tail had a single cause: reconstructing the heavy-tailed price and reserve columns produces units that are almost constant on the training rows except for spikes. For the single-field sets of PJM reserve totals a few test predictions explode and $\hat{V} = 0$ where the truth is about 0.3, and every field evaluated over that background is overestimated. Clipping predictions to the range of the target on the training rows removes this failure without using any test information. With it, reconstruction pre-

Table 7: Masked pre-training objectives over the ten targets (mean). Eff. dim.: participation ratio of the hidden representation. Clip: predictions clipped to the training range of the target. Top-3 context: share of fields whose worst background is among the three best-scanned ones. Crit. recall: scan-certify recall of τ -critical fields at $\tau = 0.5$, $\delta = 0$ (precision is 1.00 throughout). $\times 3$: predictions of three pre-training seeds averaged; the last row is the mean over three seeds of both backbones.

Backbone	eff. dim.	$V, S \leq 3$	$V, S = 4$	$M^{(2)}$	$\Gamma^{(2)}$	cert. top-1	top-3 context	crit. recall
Untrained backbone	3.30	0.0262	0.0473	0.0270	0.0240	0.884	0.65	0.92
Untrained backbone, clip	3.30	0.0253	0.0451	0.0210	0.0168	0.958	0.70	0.92
Target-only, $\times 3$ (E, main results)	1.60	0.0264	0.0379	0.0211	0.0169	0.934	0.67	0.88
Target-only, $\times 3$, clip	1.60	0.0259	0.0375	0.0214	0.0174	0.958	0.70	0.90
Reconstruction, $\times 3$	2.75	0.0200	0.0287	0.0268	0.0257	0.881	0.69	0.91
Reconstruction, $\times 3$, clip	2.75	0.0196	0.0285	0.0171	0.0150	0.961	0.71	0.91
Reconstruction $\oplus$ untrained, clip	—	0.0196	0.0295	0.0163	0.0144	0.960	0.71	0.93

training improves every metric over the target-only estimator used for the main results: the error of V by 26%, that of $M^{(2)}$ by 19% and that of $\Gamma^{(2)}$ by 11%; the certified top-1 ratio rises from 0.934 to 0.961, the worst background is among the three best-scanned ones for 71% instead of 67% of the fields, and scan-certify recovers 91% instead of 88% of the critical fields at $\tau = 0.5$ without a margin. Concatenating the features of one reconstruction backbone with those of an untrained one, as (ii) suggests, is as accurate with one trained backbone instead of three ($M^{(2)}$ error 0.016, critical-field recall 93%, means over three seeds of both backbones). Decisions move less than estimates: at $\tau = 0.7$ every estimator recovers 75–79% of the critical fields, and even the untrained backbone recovers 92% at $\tau = 0.5$. With clipping in place, target-only pre-training no longer beats the untrained backbone on any metric: its earlier advantage in certification came from being less prone to readout explosions, not from better information. The results of Sections 8–9 were computed with the target-only estimator and are, in this respect, conservative.

Warm-start fine-tuning of the whole network for each set, the natural fast alternative to retraining [25], is less accurate than the closed-form readout on nine of ten targets (on 150 random sets per data set; e.g. 0.028–0.044 versus 0.013–0.021 on RTS-GMLC) while being slower.

One pre-training for all target definitions. In practice the auditor’s target list and candidate pool change—a field is reclassified, a new target is added. With masked reconstruction over all 53 columns of a market, one backbone serves every such definition through the readout alone. Table 8 compares it with backbones supervised on each group’s own targets. The all-column backbone

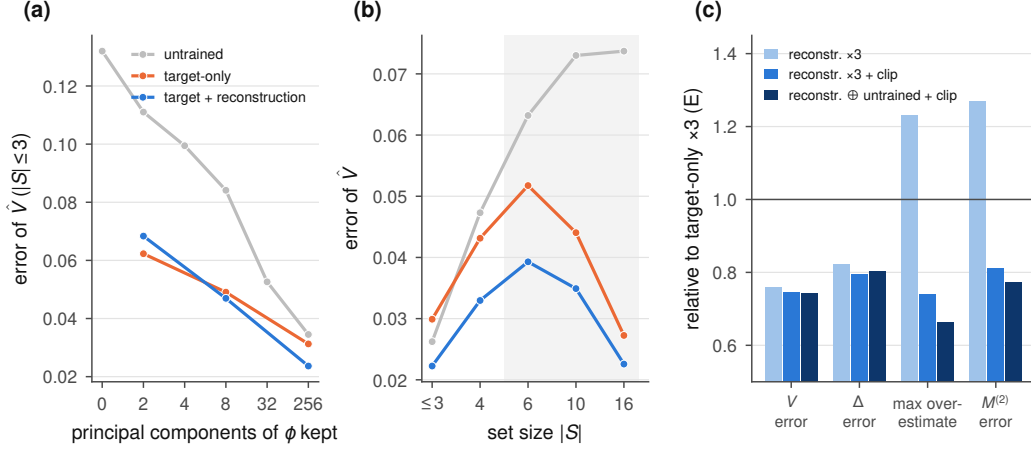


Figure 9: Why target-only masked pre-training adds little, and the repair (ten targets). (a) Error of $\hat{V}$ when the backbone features of each set are truncated to their r leading principal components (200 sets of size ≤ 3 per data set, four data sets). (b) Error of $\hat{V}$ against set size; shaded: gradient-boosting ground truth. (c) Errors of reconstruction pre-training relative to the target-only estimator E , without and with clipping predictions to the training range: set values, marginals, the largest overestimate of a marginal per field, and $M^{(2)}$.

is more accurate for V (0.040 versus 0.049) and slightly more accurate for $M^{(2)}$ (0.026 versus 0.029), and its certified top-1 ratio rises from 0.906 to 0.944.

The mechanism is visible in the representation. A backbone that serves one or two targets leaves a quarter of its hidden units inactive (24%) and has an effective dimension of 1.45; one that must reconstruct every column keeps 92% of units active and has an effective dimension of 2.44. It has to encode the latent operating state—load level, renewable output, congestion—that drives all grid quantities, which is exactly the argument for multi-task representation learning [34, 35, 36]. Why V improves more than M follows from (7): 89–98% of the V error is common-mode, and a better representation mostly shrinks this common part, which cancels in the marginals anyway.

The last row of Table 8 tests a target never seen in pre-training: the target column is deleted from the data before pre-training. Set values and marginals are estimated as well as with the all-column backbone, but $\hat{M}^{(2)}$ is inflated (bias +0.028), because the maximum over many marginals picks up the few that are overestimated; the certified top-1 ratio drops to 0.75. Retraining three backgrounds per field restores it to 0.966. Re-use for unseen targets

Table 8: Backbone scope on the four PJM/CAISO targets (same readout). Inactive: hidden units with zero variance on the training rows; eff. dim: participation ratio of the feature covariance; common-mode: $|\bar{e}|/|e|$ of the V error e .

Backbone	inactive	eff. dim	MAE V	common-mode	MAE Δ	MAE $M^{(2)}$	bias $M^{(2)}$	cert.@1	cert.@3
Per-group (targets supervised)	24.2%	1.45	0.049	98%	0.028	0.029	-0.025	0.906	0.963
All-column (masked reconstruction)	7.7%	2.44	0.040	94%	0.024	0.026	-0.021	0.944	0.976
All-column, target held out	7.5%	2.50	0.039	94%	0.024	0.056	+0.028	0.750	0.966

therefore requires certification—which is the pipeline we use anyway.

Cost. Measured on one otherwise idle GPU, retraining the attacker family sequentially for one field set takes 14–56 s (the perceptrons dominate; one network per target), whereas the amortized estimate takes 62–78 ms, a factor of 191–763. A complete $K = 2$ table (all sets of size ≤ 3) therefore takes 2–9 min to scan, against 7–53 h of sequential retraining; certification adds three retrained backgrounds per field and target, i.e. a few hundred sets. Pre-training a backbone takes about half a minute and is done once per data set. The savings compound over the recurrent workloads of an audit: re-grading after every release, redefined targets, and the sets of size four needed for $K = 3$ disposal.

11. Results V: Robustness and the Choice of K

Attacker seeds and data splits. Ground truth is itself an estimate: attackers are trained with random initializations and early-stopping splits, and V_y is measured on one random test split. We repeated the complete three-attacker retraining with two further seeds (RTS-GMLC, three targets; PJM load) and on two further random 70/30 splits (RTS-GMLC). Across these reruns the mean absolute change in V_y is 0.001–0.008, the Spearman correlation of $M^{(2)}$ across fields with the main run is 0.983–0.999, the set of τ -critical fields is identical at $\tau = 0.5$ (Jaccard 1.0 in every case) and agrees with Jaccard 0.93–1.00 at $\tau = 0.7$, and the number of combination-critical fields changes by at most one or two fields (Fig. 10c). The conclusions of Sections 8–9 are therefore not artefacts of a particular seed or split.

Threshold. Fig. 10(d) sweeps τ from 0.3 to 0.9 over all ten targets. At every threshold the share of fields that are safe alone but critical with at most two background fields exceeds the share of fields that are unsafe alone (e.g. 43% versus 10% at $\tau = 0.7$). The gap between the two grading philosophies is thus not tied to the choice of τ , although its size is (Section 12).

Choice of K . Exhaustive retraining of all sets of four fields on every data set (for NEM an additional 27,405 sets) gives the exact $M^{(3)}$. Measured by the *marginal* risk, $K = 2$ is close to saturation: $M^{(2)}/M^{(3)}$, the ratio of mean field risk, lies between 0.958 and 1.000 for nine targets and is 0.813 for NEM TUMUT3, a hydro unit whose dispatch depends on inter-regional exchange (Fig. 10a). The first background field brings most of the escalation ($M^{(1)}/M^{(2)}$ between 0.73 and 0.97).

Threshold crossings, however, do not saturate as quickly, which is the caveat stated in Section 4: a small increase of $M_{i \rightarrow y}^{(K)}$ can move a field across τ . At $\tau = 0.7$ several targets have fields that become critical only with a third background field: PJM net interchange (3 critical fields at $K = 2$, 21 at $K = 3$), NEM PPCCGT (10 and 24) and PJM load (14 and 22) (Fig. 10b). At $\tau = 0.5$ all fields are already critical at $K = 2$ except for TUMUT3 (22 versus 30). We read the two findings together: $K = 2$ is sufficient to describe and rank the risk of a field—the amplification profile of Section 8 is essentially complete at $K = 2$ —while any statement tied to a threshold is a statement about a specific budget and should be reported with it; an auditor who assumes a stronger adversary evaluates $K = 3$. Evaluating all sets of size four is exactly the workload for which the amortized scan is needed: for CAISO it comprises 52,360 sets per target, against 7,175 for $K = 2$.

12. Application: from Field Risk to Disposal

The field-level risk is a measurement; what an operator does with it depends on a threshold τ and on policy. We illustrate one downstream use and state its limits.

Critical fields. Given τ , Theorem 2 turns the risk table into a flag: field i is critical if some background of at most K fields is safe on its own and becomes unsafe when i is added. Over the ten targets the single-field rule $V_y(\{i\}) > \tau$ recovers only 13% ($\tau = 0.5$) and 17% ($\tau = 0.7$) of the critical fields. Scanning with the amortized attacker and certifying at most three candidate backgrounds per field by retraining recovers 75–88% of them with no false positives (Table 9).

Withholding. The minimal unsafe sets identify which fields to withhold. Withholding every field that is unsafe alone leaves 98% ($\tau = 0.5$) and 96% ($\tau = 0.7$) of the minimal unsafe sets of size ≤ 3 intact. Two strategies

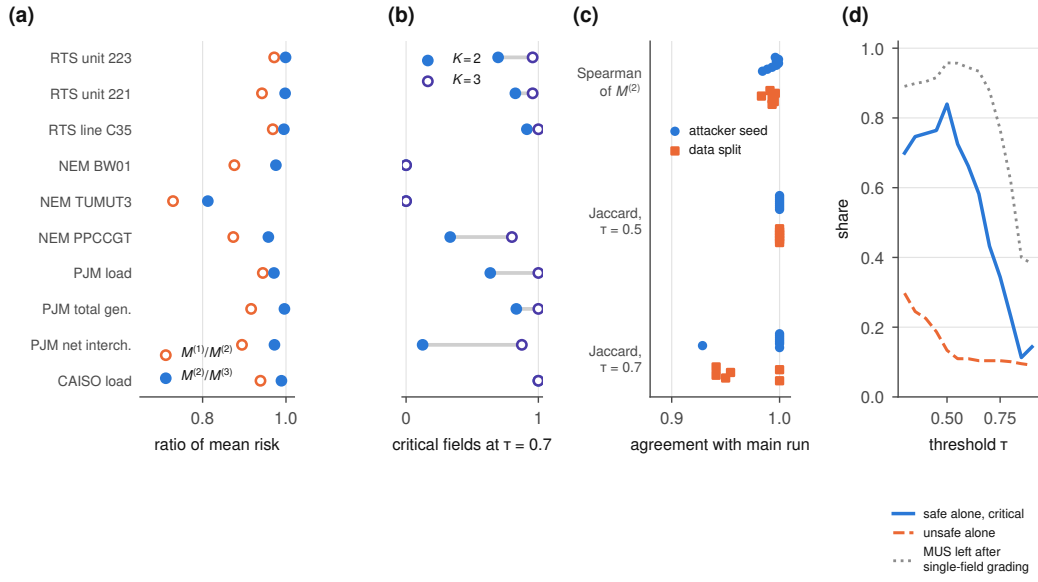


Figure 10: Robustness. (a) Saturation of mean field risk in the background budget. (b) Share of τ -critical fields at $\tau = 0.7$ with $K = 2$ (filled) and $K = 3$ (open). (c) Agreement of reruns with different attacker seeds or data splits with the main run. (d) Threshold sweep over ten targets: fields that are safe alone but critical (solid), fields unsafe alone (dashed), and minimal unsafe sets left after single-field grading (dotted).

Table 9: Decisions over the ten targets. Top: recall and precision of τ -critical fields. Bottom: fields withheld (optimum in parentheses) and share of true minimal unsafe sets left unbroken. Selection uses estimates only; retrained values are used for certification and evaluation. δ : conservative margin of the scan.

Flagging rule	$\tau = 0.5$		$\tau = 0.7$	
	recall	precision	recall	precision
Single-field rule $V(\{i\}) > \tau$	0.13	1.00	0.17	1.00
Scan on $\hat{V}$, $\delta = 0.05$	0.93	0.99	0.83	0.80
Scan + certify, $\delta = 0$	0.88	1.00	0.75	1.00
Scan + certify, $\delta = 0.05$	0.77	1.00	0.80	1.00
Withholding strategy	withheld	left	withheld	left
Single-field grading	3.3 (9.4)	98.2%	2.5 (5.1)	95.5%
Hitting set of MUS on $\hat{V}$, $\delta = 0.05$	9.4 (9.4)	5.8%	5.3 (5.1)	1.8%
Adaptive $\hat{M}^{(2)}$ greedy, $\delta = 0.05$	10.6 (9.4)	2.3%	6.1 (5.1)	1.3%

that choose fields from estimates alone—a minimum hitting set of the estimated minimal unsafe sets, and a greedy rule that withholds the field with the largest $\hat{M}^{(2)}$ and recomputes $\hat{M}^{(2)}$ over the remaining fields—leave 1.3–5.8% intact while withholding at most 1.2 fields more than the ground-truth optimum.

Limits. These numbers depend on the threshold, which is a policy choice: at $\tau = 0.5$ almost every field is critical in our data, at $\tau = 0.9$ very few are, and the share of dangerous combinations that single-field grading leaves intact varies from 96% to 38% across $\tau \in [0.5, 0.9]$. They also depend on K : minimal unsafe sets of size four exist that a withholding set computed at $K = 2$ does not break, and breaking them requires on average 11.3 instead of 9.4 withheld fields at $\tau = 0.5$ and 6.3 instead of 5.1 at $\tau = 0.7$ (Section 11). A release decision on a concrete set A can, of course, be checked directly by evaluating $V_y(A)$. We therefore present disposal as an application of the risk profile, not as its purpose: its value is to show that the per-field quantity carries enough structure—witnesses and minimal unsafe sets—to act on.

13. Discussion

What changes for grading practice. The inference risk of a power-data field is not a property of the field. It depends on the background the adversary holds,

and in our data this dependence is the rule rather than the exception: the typical field gains a third of its worst-case risk over an average background, a single background field supplies three quarters of the amplification, and the most amplified fields—prices, renewable forecasts, available capacity—are exactly those a market must publish. Single-field grading is therefore not a conservative approximation of a background-aware audit; it errs in the unsafe direction. The profile proposed here keeps the familiar form of a grade per field but reports what the grade depends on: the single-field risk $M^{(0)}$, the amplification $\Gamma^{(K)}$, and the family of backgrounds that attain it. In the controlled mechanism these backgrounds are the physically correct ones—the unit output for every price, and the own-bus price before the remote one, except when the line between them is congested. Because $M_{i \rightarrow y}^{(K)}$ is defined relative to the public baseline, the profile is recomputed whenever something is released; the amortized attacker makes this cheap and certification keeps it honest.

Relation to inference-graph approaches. Propagating risk over pairwise inference relations [6] captures chains of single-field inferences and physical relations known to experts, and it does not require an attacker to be trained. It cannot represent joint inference by construction (Section 7), and its scores are not checkable by an attack experiment. Set-level inferability provides the missing ground truth; an inference graph could be used to propose candidate backgrounds for the scan, a combination we have not evaluated.

What the estimator is and is not. The amortized attacker is not required for correctness: every decision reported here is certified by retraining. Its role is to make the scan cheap enough to be repeated, and its accuracy matters only through the ranking of backgrounds that certification follows. Most of its accuracy comes from solving a readout per field set; on sets of a few fields, a random feature map already supplies most of what the readout needs, which is why target-only masked pre-training—whose representation collapses onto the targets—adds little. Pre-training earns its place when it has to reconstruct all columns: the representation then encodes the operating state, the estimates improve on every metric, and the advantage grows with the size of the queried sets. A maximum over many estimated marginals is sensitive to rare readout failures rather than to average accuracy, so safeguards against exploding predictions matter more for $M_{i \rightarrow y}^{(K)}$ than for V_y .

Limitations. (i) Risk is defined relative to a fixed attacker family; a stronger attacker can only raise V_y and hence the measured combination risk, but the numerical values are family-dependent. (ii) The adversary is assumed to hold labelled history of the target; with little auxiliary data, $M_{i \rightarrow y}^{(K)}$ as a maximum statistic is inflated by selection noise, which again argues for certified lower bounds rather than point values. (iii) We use random train/test splits; drift over time may change the values of $M_{i \rightarrow y}^{(K)}$, although not the audit procedure. (iv) $K = 2$ nearly saturates the marginal risk on nine of ten targets but not threshold crossings, so threshold-based statements carry their budget; parity-like targets would defeat any fixed K . (v) Field roles for NEM, PJM and CAISO follow the Chinese disclosure rules rather than local market rules; they are used to define a realistic audit, not to assess those markets. (vi) The mechanism case has one unit and one private parameter; it validates that the definitions recover a known structure, not that every amplification in real data has a simple physical reading. (vii) The inference-graph baseline reimplements the propagation idea without the expert inputs and attention training of [6].

14. Conclusion

We defined field-level inference risk from set-level inferability, the out-of-sample accuracy of retrained attackers on a set of fields. The budgeted marginal risk $M_{i \rightarrow y}^{(K)}$ is the worst-case gain that releasing a field brings over any background of at most K fields, relative to what is already public, and its excess over single-field risk is the background amplification $\Gamma^{(K)}$; $M_{i \rightarrow y}^{(K)}$ is a threshold-free safety margin, and its critical fields are exactly the members of small minimal unsafe sets. An amortized attacker—masked pre-training with a closed-form readout per field set—scans all small sets in milliseconds, and retraining a few scanned backgrounds per field certifies the result. On four power data sets with rule-based field roles and exhaustive retraining as ground truth, and on a controlled market case with a known mechanism, background amplification proved common, physically interpretable and invisible to single-field and attribution scores: single-field grading recovered 13–17% of the critical fields, whereas the scan–certify pipeline recovered 75–88% of them with no false positives. The profile of single-field risk, amplification and dangerous backgrounds is the per-field quantity a classification standard asks for when it asks how aggregation changes a field’s grade.

Data and code availability

RTS-GMLC [39] and the AEMO NEMWEB archive [40] are public; the PJM and CAISO series were obtained from the operators’ public data portals. Field-role tables, the code for dispatch simulation, ground-truth retraining, the amortized attacker and all analyses will be made available upon publication.

Appendix A. Further properties of $M_{i \rightarrow y}^{(K)}$

Proposition 3 (Ladder and candidate-set monotonicity). (a) $M_{i \rightarrow y}^{(0)} \leq M_{i \rightarrow y}^{(1)} \leq \dots \leq M_{i \rightarrow y}^{(|H|-1)}$; (b) *enlarging the candidate set from H to $H' \supseteq H$ cannot decrease $M_{i \rightarrow y}^{(K)}$* ; (c) *if f' duplicates f , i.e. $V_y(B \cup T \cup \{f'\}) = V_y(B \cup T \cup \{f\}) = V_y(B \cup T \cup \{f, f'\})$ for all T , deleting f' leaves $M_{i \rightarrow y}^{(K)}$ unchanged for every $i \neq f, f'$.*

Proof. (a) and (b) are maxima over nested feasible sets. (c) If a witness T of i contains f' , replace f' by f (if $f \notin T$) or drop it (if $f \in T$); the value is unchanged and the budget is respected, so the maximum does not decrease after deleting f' , and by (b) it does not increase. $\square$

Theorem 3 (Truncated super-efficiency). *For $A \subseteq H$ with $|A| \leq K + 1$, $V_y(B \cup A) - V_y(B) \leq \sum_{i \in A} M_{i \rightarrow y}^{(K)}$. More tightly, with $U^{(K)}(\emptyset) = 0$ and $U^{(K)}(A) = \min_{j \in A} [U^{(K)}(A \setminus \{j\}) + M_{j \rightarrow y}^{(\min(|A|-1, K))}]$, $V_y(B \cup A) - V_y(B) \leq U^{(K)}(A) \leq \sum_{i \in A} M_{i \rightarrow y}^{(K)}$.*

Proof. Order $A = \{i_1, \dots, i_m\}$ and telescope $v_B(A) = \sum_t [v_B(A_t) - v_B(A_{t-1})]$; the t -th term is a marginal of i_t over a background of size $t - 1 \leq K$, hence at most $M_{i_t \rightarrow y}^{(K)}$. The staged bound follows by induction on $|A|$ and Proposition 3(a). $\square$

For $K \geq |H| - 1$ this is the super-efficiency of MCI [7]; with a finite budget it is stronger because $M_{i \rightarrow y}^{(K)} \leq \text{MCI}$. It yields a release rule “accept A if $U^{(K)}(A) \leq \tau$ ” that never accepts an unsafe set of size $\leq K + 1$. We do not use this rule in the main text because it is conservative for larger release sets; the minimum-hitting-set disposal of Section 9 addresses the same need directly.

Proposition 4 (Irreducible synergy is bounded by field risk). *With $\text{syn}^B(S) = V_y(B \cup S) - \max_{T \subsetneq S} V_y(B \cup T)$ and $|S| \geq 2$, $\text{syn}^B(S) \leq \min_{j \in S} M_{j \rightarrow y}^{(|S|-1)}$.*

Proof. $\max_{T \subsetneq S} V_y(B \cup T) \geq V_y(B \cup S \setminus \{j\})$ for every j , hence $\text{syn}^B(S) \leq \Delta_j(B \cup S \setminus \{j\}) \leq M_{j \rightarrow y}^{(|S|-1)}$. $\square$

Remark 1 (Escalation is weaker than synergy). $M_{i \rightarrow y}^{(K)} = M_{i \rightarrow y}^{(0)}$ iff $\Delta_i(B \cup T) \leq \Delta_i(B)$ for all $|T| \leq K$. Submodularity implies no escalation but not conversely: with $V_y(\{1\}) = 0.8$, $V_y(\{2\}) = 0.6$, $V_y(\{3\}) = 0.8$, all pairs 0.8 and $V_y(\{1, 2, 3\}) = 1$, every field has $M^{(2)} = M^{(0)}$, yet $\Delta_1(\{3\}) = 0 < \Delta_1(\{2, 3\}) = 0.2$.

Appendix B. Implementation details

Attackers. Single- and multi-target perceptrons: two hidden layers of 128 ReLU units, dropout 0.15, Adam (learning rate 10^{-3} , weight decay 5×10^{-4}), batch 128, at most 300 epochs, patience 60, epoch chosen per (set, target) on validation. Many attackers are trained in parallel on one GPU as independent networks whose inputs are masked to their own field set; masked columns receive no gradient, so each network is functionally a network trained on its set alone. Gradient boosting: histogram gradient boosting with (learning rate, leaves) $\in \{(0.05, 15), (0.1, 31)\}$, 300 iterations with internal early stopping; the configuration is chosen on validation. The family member is chosen per (set, target) on validation. Across data sets gradient boosting is selected for 35–76% of (set, target) pairs (RTS-GMLC 76%, NEM 35%, PJM 67–69%, CAISO 52%), so the family maximum matters.

Estimator. Backbone: three hidden layers of 256 ReLU units with dropout 0.1, input $[x \odot m, m]$, Adam (10^{-3} , weight decay 5×10^{-4}), batch 256, at most 400 epochs, patience 60 on eight fixed validation masks. Ridge penalties $\lambda \in \{10^{-3}, \dots, 10^2\}$.

Inference-graph baseline. Following the structure of [6] without its expert-assigned intrinsic scores: nodes are the candidate fields and the target; the weight of edge $i \rightarrow j$ is the validation R^2 of predicting field j from field i alone by gradient boosting; the score of i is $\sum_{h=1}^3 0.5^{h-1} (W^h)_{i,y}$. The first term equals single-field predictability; longer paths add inference chains through other fields.

Table B.10: Field roles and rule basis (Chinese electricity-market disclosure rules [1]; appendix item numbers in parentheses).

Data set	Candidate fields (released before the target)	Targets
RTS-GMLC	zonal load forecasts (3; 6.41); wind, PV, rooftop PV and hydro forecasts (4; 6.42); regulation up/down and spinning reserve requirements (3; 6.46); system energy price and nodal prices at 9 buses (10; 6.52); congestion indicators of lines C6, C29, CA-1 (3; 6.54)	outputs of units 223_STEAM_3 and 221_CC_1 (specific information, Art. 17(3)); hourly flow on line C35 (only daily averages of important lines are listed, 6.66)
NEM 2024	regional demand (5), available capacity (5), semi-scheduled wind and solar forecasts (9), regional prices (5), interconnector flows (6)	actual outputs of units BW01, TUMUT3, PPCCGT (specific information)
PJM	regulation market prices and volumes (5); day-ahead reserve prices, requirements, self-schedules and cleared volumes (9); day-ahead system energy price, LMP, congestion and loss components (4); real-time LMP, congestion and loss components (3); scheduled interchange (1); for total generation and net interchange also the system load forecasts (2)	metered load (6.61), total generation (6.57), net actual interchange (6.65), all released next day
CAISO	day-ahead load forecasts of three sub-areas (3); day-ahead wind and solar forecasts (6); day-ahead and 15-min real-time prices and components (11); day-ahead schedules of sub-area load, total generation, imports and exports (6); ancillary-service requirements incl. regulation mileage (5) and procured volumes (4)	actual CAISO load (6.61), released next day

SAGE. Marginal imputation with 32 background rows, 512 evaluation rows from the validation split and 24 random permutations, on a gradient-boosting model fitted to all candidates.

Appendix C. A synthetic example

Fig. C.11 uses synthetic data with a known structure, $Y = aAB + bP + \sigma\epsilon$, where A and B act only jointly, C_1 and C_2 are two noisy copies of P , D is a weak copy of P and N is noise; all 63 sets are retrained. Correlation, single-field retraining and pairwise inference-graph propagation give A and B a score of zero although $\{A, B\}$ reaches 0.54. LOCO gives C_1 and C_2 almost zero because each substitutes for the other, and LOCO, permutation importance and *SAGE* all give D zero because the full model relies on C_1, C_2 instead—yet $\{A, B, D\}$ crosses $\tau = 0.7$ while $\{A, B\}$ does not, so D is critical. *SAGE* splits credit within both pairs. $M^{(2)}$ assigns A, B, C_1, C_2 their full worst-case contribution and gives D a non-zero risk with the witness $\{A, B\}$. This is the behaviour predicted by Table 1: averages dilute, single contexts miss substitutes, and pairwise edges miss joint inference.

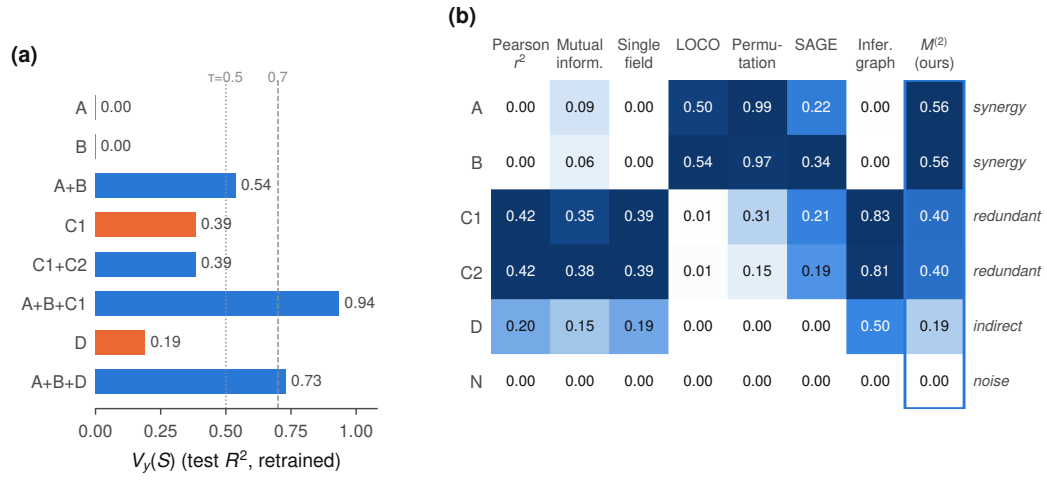


Figure C.11: Synthetic mechanism example. (a) Retrained inferability of selected sets. (b) Scores of eight methods (colour normalized per column); the last column is $M^{(2)}$.

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